

# SAGAR-SAKSHI · EVIDENCE CARD

Smart India Hackathon 2026 | Team Tantragyan (SIH26143) | Run ID: run\_kerala\_20260924\_074715

|                         |                                                                  |
|-------------------------|------------------------------------------------------------------|
| EVIDENCE GRADE          | GRADE None                                                       |
| HEADLINE ASSESSMENT     | Insufficient evidence                                            |
| AIS DATA STATUS         | SYNTHETIC (Validated replay log)                                 |
| MANIFEST HASH (SHA-256) | 3955ca80de8a4a51423d654581ab83af25d72080ef7e3c6a6dc59d45c18f6b6a |

## Candidate Hypotheses & Posterior Probabilities P(H|E)

| Hypothesis / Source       | Posterior P(H E) | Drift Fit | Head Prox. | Behavior |
|---------------------------|------------------|-----------|------------|----------|
| Null / Unknown Source (U) | 0.8996 (90.0%)   | N/A       | N/A        | N/A      |
| MSC ELSA 3                | 0.0952 (9.5%)    | 0.65      | 0.95       | 0.30     |
| DARK_sar_ship_001         | 0.0006 (0.1%)    | 0.05      | 0.05       | 0.30     |
| DARK_sar_ship_002         | 0.0006 (0.1%)    | 0.05      | 0.05       | 0.30     |
| DARK_sar_ship_003         | 0.0006 (0.1%)    | 0.05      | 0.05       | 0.30     |
| DARK_sar_ship_004         | 0.0006 (0.1%)    | 0.05      | 0.05       | 0.30     |
| DARK_sar_ship_005         | 0.0006 (0.1%)    | 0.05      | 0.05       | 0.30     |
| DARK_sar_ship_006         | 0.0006 (0.1%)    | 0.05      | 0.05       | 0.30     |
| DARK_sar_ship_007         | 0.0006 (0.1%)    | 0.05      | 0.05       | 0.30     |
| DARK_sar_ship_008         | 0.0006 (0.1%)    | 0.05      | 0.05       | 0.30     |
| OCEAN PRIDE               | 0.0002 (0.0%)    | 0.02      | 0.02       | 0.30     |

## Exculpatory Evidence: Excluded Vessels

| Vessel ID     | Exclusion Reason                                                               |
|---------------|--------------------------------------------------------------------------------|
| KERALA STAR   | Track never entered backward reachability envelope (closest approach: 15.0 nm) |
| MALABAR QUEEN | Track never entered backward reachability envelope (closest approach: 6.9 nm)  |

## Operational Caveats & Legal Disclaimers

- Non-Accusatory Standard:** SAGAR-SAKSHI produces investigative leads based on multi-sensor evidence consistency. It does not establish legal liability or criminal culpability.
- Detectability Mask:** SAR oil detection depends on Bragg scattering in 3-10 m/s wind regimes.
- Radar Limits:** Sentinel-1 C-band SAR resolution limits reliable vessel detection to craft >= 20m.